

PCC - KOICA · NATIONAL DAIRY HERD IMPROVEMENT PROJECT · LOCAL
TRAINING SERIES

Smart Dairy Herd Management

Reproduction, Health & Sensor-Based Monitoring

Korea's dairy experience applied to Philippine dairy cattle and
buffalo

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Science City of Muñoz, Nueva Ecija · 2026



Your lecturer and the farm behind this lecture

Dr. Hyunje Ha, DVM

- 30+ years large-animal clinical veterinarian
- 18+ years embryo transfer / IVF — ~3,000 transfers per year (Genetics Co.)
- Dairy farmer: Song Young Shin Farm, Anseong (founded 2011) — A2 Jersey hay milk
- First in Korea to run smaXtec rumen sensors on a commercial dairy
- Livestock-environment technology (D2O bedding, compost, odor & carbon)
- Lecturer, KOICA dairy-farmer training programs; Konkuk University adjunct professor

Song Young Shin Farm — a working test bed

120

head (65 milking + 55 heifers)

2

robotic milking units

90

rumen bolus sensors in use

3,000 m²

humus bedded-pack barn

1st

animal-welfare certified dairy
in Korea

1st

low-carbon certified dairy in
Korea

AGENDA

Today's programme



Module 0 · Korea's dairy journey

From a 1969 demonstration farm to 10,000 kg cows — what actually moved the needle

30 min



Module 1 · Reproduction management

Estrus, AI timing, pregnancy diagnosis, calving, colostrum, genetics tools, KPIs

90 min



Module 2 · Health, mastitis & biosecurity

Transition cows, mastitis detection & treatment, heat stress, records and drug use

90 min



Module 3 · Digital herd monitoring

Rumen bolus sensors: how they work, alerts, cases, NDHIS/iHealth integration, pilot design

90 min



Module ★ · Practicum stations

Heat detection & AI timing · ultrasound PD · CMT & milking routine · bolus demo & app

120 min

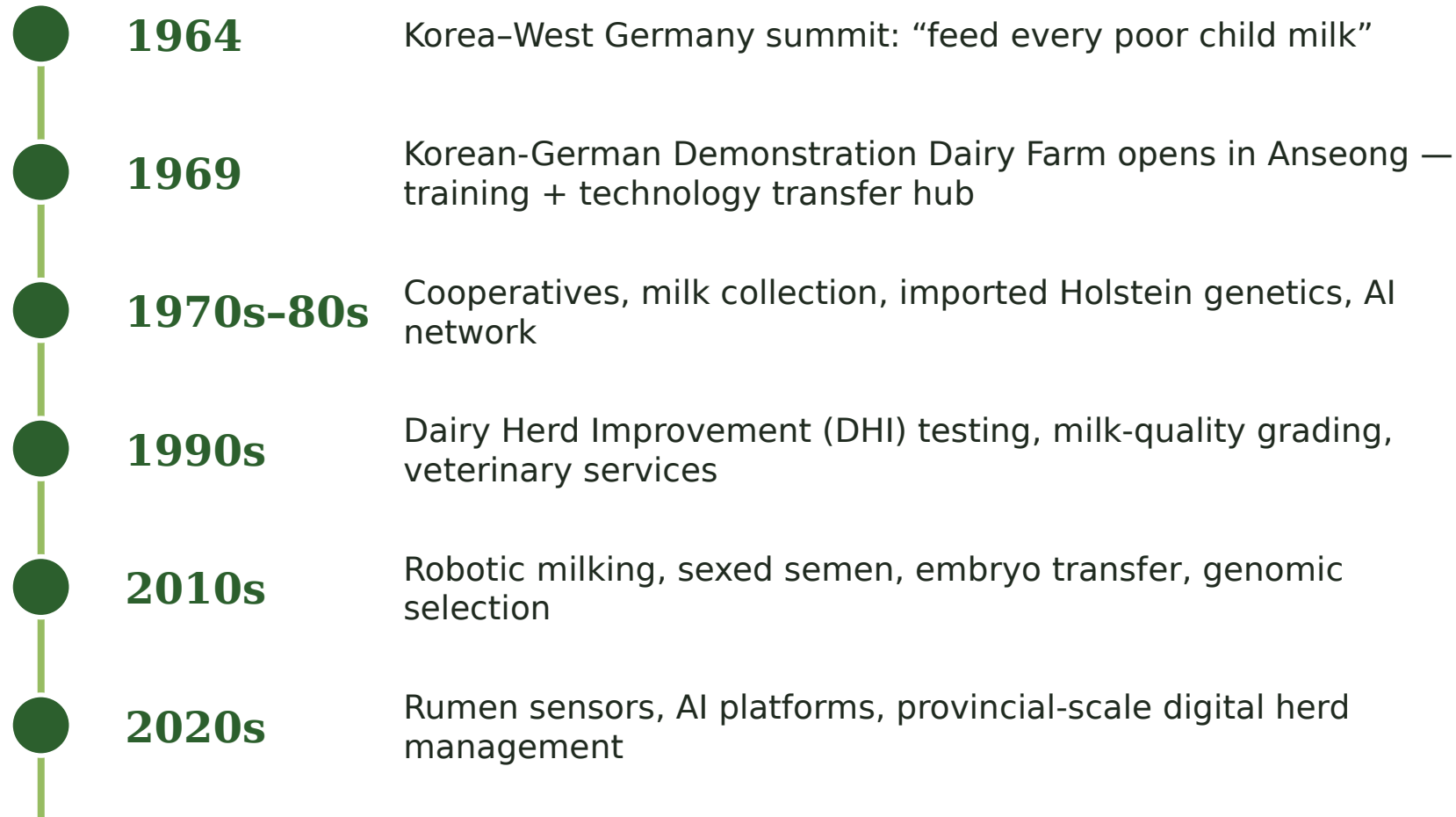


MODULE 0

Korea's dairy journey

Fifty years from a demonstration farm to one of the world's highest-yielding national herds — and why the same playbook fits the Philippines.

It started with one demonstration farm

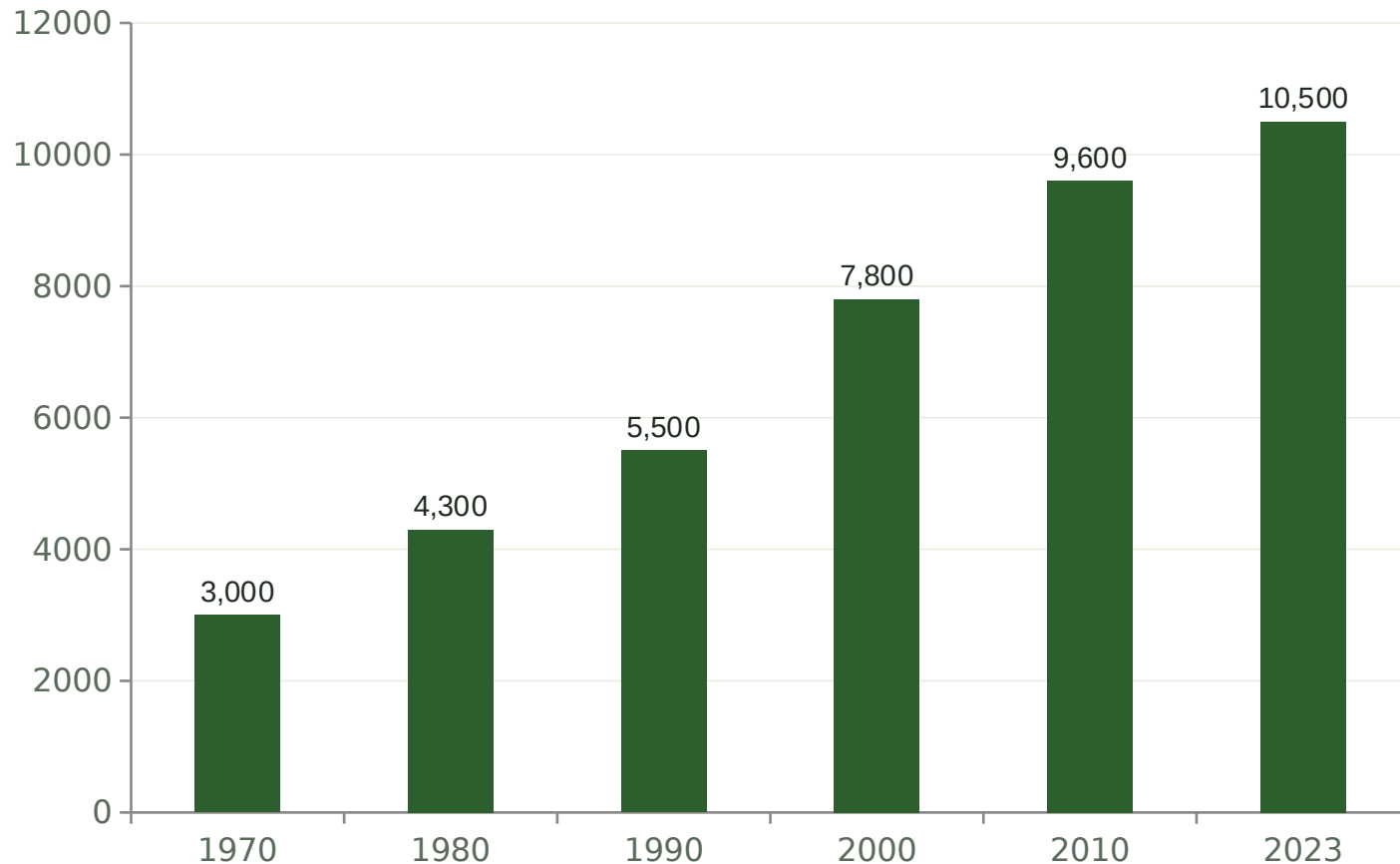


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The Philippines today is where Korea was in 1969: a small herd, imported know-how, and a government that decided dairy matters. The difference now is that data tools exist from day one.

What five decades of management built

Korea — average milk yield per cow (kg / year, approximate)



Philippine starting point

- Local milk supplies only a small share of national demand (roughly 1–2%) — huge room to grow
- Smallholder dairy buffalo: about 4–8 L/day; improved management and genetics push this up
- Dairy cattle herds in Luzon and Mindanao reach 15–20 L/day with good feeding
- Korea started at ~3,000 kg/cow — yield tripled through genetics, feeding, health and records

Korean figures approximate, from national dairy statistics; Philippine figures indicative — confirm with PCC/PSA data.

Four things that actually moved the needle



Genetics

AI with proven sires, then sexed semen, embryo transfer and genomic selection. Gains compound every generation.



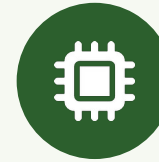
Records & herd improvement

Individual-animal IDs, DHI milk testing, reproduction and health events written down and analysed — the ancestor of NDHIS.



Reproduction & health

A calf a year, mastitis under control, transition cows protected. Technicians and vets working from the same records.



Data & sensors

Rumen bolus sensors, robots and AI turn observation into 24/7 measurement. The last mile, but built on the first three.

Same playbook for NDHIP: identify every animal → record every event → manage reproduction and health from the records → add sensors to see what eyes cannot.



MODULE 1

Reproduction management

Heat detection, AI timing, pregnancy diagnosis, calving, colostrum, genetic tools and the KPIs that tell you whether it is working.

Reproduction is the engine of the dairy business

1 calf

per cow per year is the goal
— every lactation starts with a calving

12-13 mo

target calving interval for dairy cattle (buffalo: aim $\leq$ 15-16 months)

≤ 120 d

target days open (calving to conception) in cattle

Every day

open beyond target costs feed, lost milk and a later calf

The reproduction chain — and where it usually breaks

Cow cycles after calving

⚠ Uterine infection, negative energy balance

Heat is detected

⚠ Silent / night heats — the #1 loss

AI at the right time

⚠ Too early / too late, poor semen handling

Conception & pregnancy confirmed

⚠ Open cows found too late

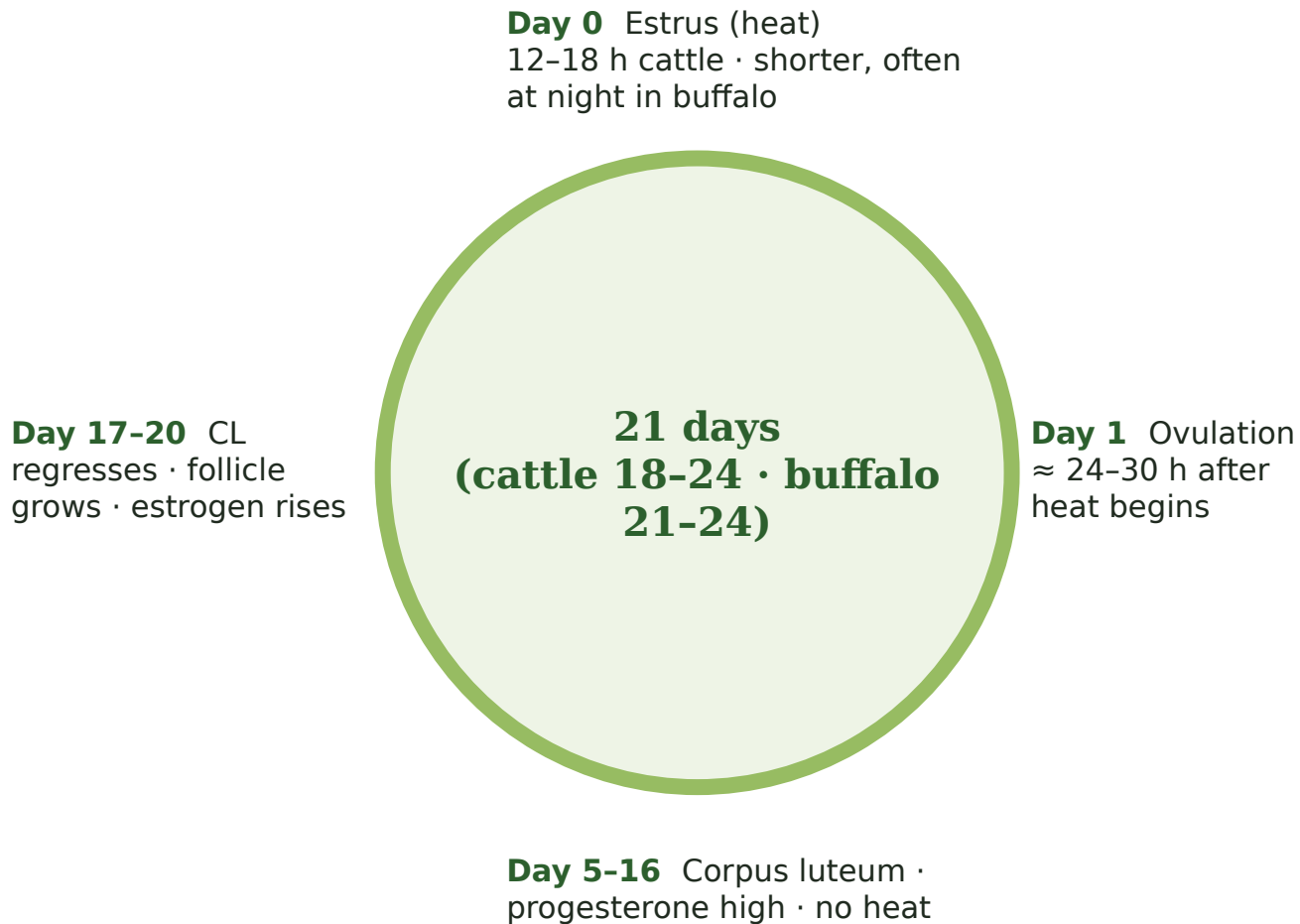
Healthy calving

⚠ Dystocia, milk fever

Calf survives

⚠ No colostrum, scours

The estrous cycle — the clock you are working with



Practical rules that follow from the clock

- First heat after calving: cattle 2–4 weeks, but wait until ≥ 45 –60 days (voluntary waiting period) before breeding
- A cow not seen in heat by 60 days post-calving needs a check: ovaries, uterus, body condition
- Buffalo: seasonal tendency (more cycling in cooler months), weaker outward signs — plan for aids, not eyes alone
- Heat signs last hours, not days — observation 3× daily (early morning, evening, night) or continuous sensing
- If you know the heat date, you know the next heat date (+21 d) and the pregnancy-check date

Heat detection — the number one leak in the bucket

Sign	Reliability	Notes
Standing to be mounted	Primary (definitive)	Cattle: lasts 8–18 h · Buffalo: brief, often at night
Mounting other animals, restlessness, bellowing	Secondary	Occurs before and during standing heat
Clear stringy vulvar mucus, swollen vulva	Secondary	Buffalo: mucus is often the most visible sign
Drop in milk yield and feed intake, rise in activity	Secondary	Exactly what sensors measure automatically
Chin resting, sniffing, rubbed tailhead	Supportive	Use tail paint / chalk to read overnight mounting

Target: heat detection rate above 60% of eligible animals per 21 days. Below 40%, no AI programme can succeed.

Silent heat: a normal ovulation with no visible signs. Reported in a large share of buffalo and of high-yielding cows — which is why Korea moved to sensors.

Detection aids — from cheap to continuous



Tail paint / chalk — reads mounting you did not see



3 × 30 min observation, incl. night — free, but labour-bound



Pedometers / collars — activity only, loss risk

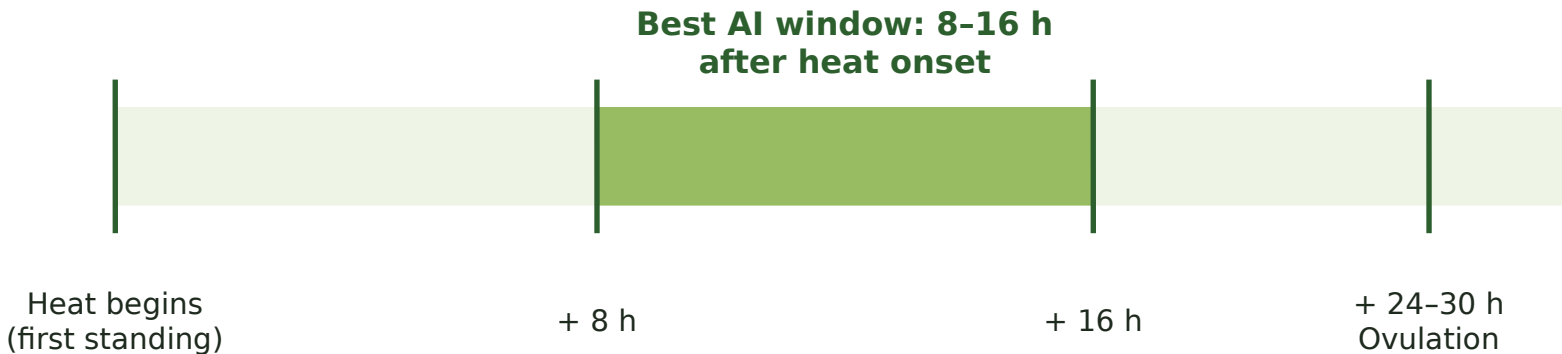


Rumen bolus — activity + temperature + rumination, inside the animal



Synchronisation (Ovsynch, PGF) — timed AI when detection is weak

AI timing and semen handling



AM / PM rule

- Seen in heat in the morning → inseminate that evening
- Seen in the evening → inseminate next morning
- Once-a-day AI: inseminate all animals seen in heat that day (works well in practice)
- Sensor heat alert: it gives the onset time and an “optimal insemination window”

Semen handling

- Keep straws below the frost line; lift canister ≤ 10 s
- Thaw at 35–37 °C for 30–45 s, dry the straw
- Protect from cold shock and sunlight; use within 15 min
- Deposit in the uterine body, gentle, clean sheath and gun
- Sexed semen: thaw exactly as instructed, inseminate closer to ovulation

Common reasons for AI failure

- Inseminating a cow that was not truly in heat (≈ 10 –20% of AIs in weak-detection herds)
- Timing too early (first sign) or too late (next day)
- Semen damaged by poor thawing or repeated exposure
- Uterine infection not cleared before breeding
- Heat stress (THI > 72) — embryo loss in the first week
- Poor body condition, negative energy balance

Pregnancy diagnosis — find the open cow early

Method	Earliest	Accuracy	Practical notes
Non-return to heat	Day 18-24	Low (misses silent heats)	Free, but only a screen; sensors improve it
Ultrasound (rectal)	Day 28-30	High; sees embryo & heartbeat	Also detects twins, uterine fluid, ovarian cysts — the technician's best tool
Rectal palpation	Day 35-45	High in trained hands	No equipment; buffalo rectum is narrower — gentle technique
PAG blood / milk test	Day 28+	High	Lab or cow-side kit; good where ultrasound is scarce
Milk progesterone	Day 21	High for “open”, weaker for “pregnant”	Best used to confirm an open animal quickly



Check at day 28-35 with ultrasound, re-check at day 60-90 to catch embryo loss



Open animal → re-enrol immediately: observe / synchronise, do not wait for “natural” heat



Enter every result in NDHIS — pregnancy rate is the KPI the whole system is built on

Calving — prepare, watch, intervene at the right time

Signs calving is near

- Udder fills, teats waxy (days)
- Pelvic ligaments relax, tailhead sinks (24–48 h)
- Restlessness, isolation, reduced feed intake
- Vulva swollen, mucus plug passes
- Body temperature drops $\approx 0.3\text{--}0.5\text{ }^{\circ}\text{C}$ in the last 24 h — what the bolus detects
- Move to a clean, dry calving pen with water

Stages of labour & time limits

Stage 1

2–6 h (heifers up to 12 h)

Cervix dilates, restlessness. Do not disturb.

Stage 2

Cow $\leq 2\text{ h}$ · heifer $\leq 3\text{ h}$

Water bag $\rightarrow$ feet $\rightarrow$ calf. No progress for 30 min after water bag = examine.

Stage 3

Placenta within 12 h

Retained $> 24\text{ h}$ = retained placenta, watch for metritis.



Call the vet when

- Only head or only one leg, or tail first
- Calf too large, no progress after 20–30 min of traction
- Uterine torsion (no water bag, twisted vagina)
- Cow down before or after calving (milk fever)
- Bleeding, prolapse, foul discharge

Colostrum and the first week decide next year's herd

Q Quality

IgG > 50 g/L — Brix refractometer $\geq$ 22%. Discard bloody or mastitic colostrum.

Q Quantity

10% of body weight in the first day: 3–4 L for a 35–40 kg calf, 2–3 L for a buffalo calf.

Q Quickly

First feed within 2 h of birth, second by 6–8 h. Gut absorption falls sharply after 12 h.

S sQueaky clean

Clean udder, clean bottle or tube, refrigerate $\leq$ 24 h or freeze. Bacteria block IgG uptake.



Dip in 7% iodine or chlorhexidine at birth and again at 12 h



Dry bedding, shade, ventilation without draughts; individual or small groups



Fluids first (oral electrolytes 2–4 L/day), keep feeding milk, isolate; antibiotics only for systemic signs



Pre-weaning mortality < 5%; double birth weight by weaning; record every treatment

Genetic tools: AI, sexed semen, embryos, genomics



Artificial insemination

Proven sires, standardised handling. The foundation — 100% of Korean dairy replacements come from AI or ET.



Sexed semen

≈ 90% female calves. Use on heifers and best cows, inseminate slightly later; expect somewhat lower conception than conventional.



Embryo transfer (IVF / OPU)

One elite donor → many calves. Genetics Co.: ~3,000 transfers/yr, ~60% conception in recipients with good management.



Genomic selection

DNA test of heifer calves predicts merit before first calving. Select replacements early, cull low-merit animals sooner.

Relevance for the Philippines

- Dairy cattle: imported sexed Holstein embryos in local recipient cows are a proven route to build a high-merit herd without importing live animals (Korea-Uzbekistan programme with RDA as reference).
- Dairy buffalo: PCC's own AI, ET and genomic programmes apply — sensors add the heat-detection precision these tools depend on.

Reproduction KPIs — what NDHIS should show monthly

KPI	Definition	Target (dairy cattle)	Buffalo (indicative)	Where it comes from
Heat detection rate	Eligible animals bred ÷ eligible in 21 d	> 60%	> 50%	Heat & AI records (or sensor alerts)
Conception rate	Pregnant ÷ inseminated	> 40%	> 45%	AI + pregnancy diagnosis records
21-day pregnancy rate	Detection rate × conception rate	> 20%	> 18%	Calculated in NDHIS
Calving-to-first-service	Days from calving to first AI	< 75 d	< 90 d	Calving + AI records
Days open	Calving to conception	< 120 d	< 150 d	Calving + PD records
Calving interval	Calving to next calving	< 400 d	< 480 d	Calving records
Calf mortality (pre-weaning)	Deaths ÷ born alive	< 5%	< 5%	Calf & iHealth records

Targets are management goals for well-run herds, not averages. Set each farm's baseline first, then improve one KPI at a time — usually heat detection rate first, because it moves everything else.

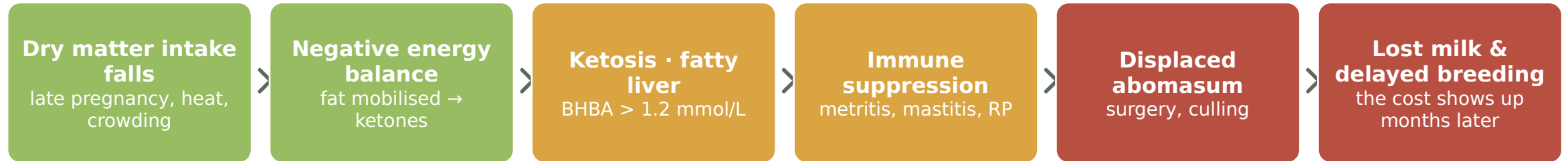


MODULE 2

Health, mastitis & biosecurity

The transition period, metabolic disease, mastitis detection and treatment, heat stress in the tropics, biosecurity, records and responsible drug use.

The transition period — the fresh-cow danger zone



Milk fever (hypocalcaemia)

- Signs: cold ears, muscle tremor, S-shaped neck, down within 72 h of calving; subclinical cases are 5× more common
- Treatment: IV calcium slowly (monitor heart), oral calcium bolus for standing cows
- Prevention: low-calcium or anionic (DCAD) dry-cow ration, avoid over-fat dry cows (BCS 3.0–3.5), oral calcium at calving for older cows

Ketosis

- Signs: reduced appetite (grain first), rapid weight loss, sweet breath, milk drop in week 1–4; often silent
- Cow-side test: blood BHBA ≥ 1.2 mmol/L (subclinical), ≥ 3.0 (clinical); milk/urine ketone strips
- Treatment: oral propylene glycol 300 mL/day for 3–5 days, IV glucose for clinical cases, treat the cause (feed access, other disease)
- Prevention: maximise DMI in the last 3 weeks, fresh feed and water, low stocking density, monitor rumination

Mastitis — the most expensive disease on a dairy farm

	Contagious mastitis	Environmental mastitis
Main pathogens	Staph. aureus, Strep. agalactiae, Mycoplasma	E. coli, Strep. uberis, Klebsiella, environmental staph
Source	Infected udders → spread at milking (hands, liners, cloths)	Bedding, mud, manure, water; between milkings and in the dry period
Typical picture	Chronic, subclinical, high SCC, low cure rate	Acute clinical cases, sometimes severe (E. coli), fresh cows
Control focus	Milking hygiene, post-dip, segregation, cull chronic cows	Clean dry bedding, pre-dip, dry-cow therapy, teat sealants

Buffalo milk: high fat and SCC baseline differ from cattle — use CMT trends within the animal rather than cattle cut-offs.

Numbers to know

< 200,000

cells/mL — bulk tank target

> 200,000

cells/mL in a cow = subclinical infection

≈ 5-10%

of yield lost per doubling of SCC above 100k

30-50%

of the cost is discarded milk + lost yield; treatment is the smaller part

Detect early, treat correctly, record everything

California Mastitis Test (CMT) — 4 steps

- 1 Strip 2–3 squirts, then 2 mL milk per cup
- 2 Add equal volume of CMT reagent
- 3 Swirl 10 s, hold horizontal
- 4 Read: no gel = negative · slight gel = trace · thick gel = 2–3 (infected)

Clinical grades

Grade	Signs	Action
1 Mild	Abnormal milk only	Strip out, monitor, culture if possible, NSAID; antibiotic only per protocol
2 Moderate	Milk + swollen, hot quarter	Intramammary antibiotic per vet protocol + NSAID, frequent stripping
3 Severe	Sick cow: fever, off feed, down	Vet now — systemic antibiotics, fluids (E. coli), anti-inflammatory

Responsible antimicrobial use

- Diagnose before treating: CMT, then culture where a lab exists — many E. coli cases self-cure without antibiotics
- Use the farm's written treatment protocol signed by the veterinarian
- Full course, correct dose by body weight, correct route; never “one tube and stop”
- Respect milk and meat withdrawal periods — mark the cow, milk her last, record dates
- Dry-cow therapy: selective, based on SCC/CMT history; teat sealant for the rest
- Korea's experience: early detection by rumination and temperature alerts reduced antibiotic use markedly (manufacturer reports up to 70%)

A ten-step milking routine prevents mastitis



1

Clean hands, gloves



2

Check udder, strip 2-3
squirts into a cup



3

Pre-dip 30 s



4

Wipe dry — one towel
per cow



5

Attach within 60-90 s
of stimulation



6

Adjust cluster; no air
leaks



7

Detach when flow
drops — avoid over-
milking



8

Post-dip every teat, full
coverage



9

Milking order: fresh &
healthy → high-SCC →
treated last



10

Rinse, wash (hot +
detergent), acid rinse,
dry equipment

Hand milking (smallholder buffalo): same principles — clean hands, dry teats, full milking, post-dip, milk sick animals last.

Heat stress — the tropical dairy's permanent challenge

THI	Stress level	What happens to the animal
< 68	Comfort	Normal intake, rumination and fertility
68–71	Mild	Breathing > 60/min, seeks shade, intake begins to fall
72–79	Moderate	Milk –10 to –25%, conception rate falls sharply, early embryo loss
80–89	Severe	Panting, drooling, standing, rumination stops; risk to life above 90

THI = temperature-humidity index. At 30 °C and 80% RH the THI is $\approx$ 84 — a normal afternoon in Nueva Ecija.

What the bolus adds

- Core body temperature per animal, not just barn THI — shows which animals are actually suffering
- Water-intake events and rumination drop — the earliest signs; heat-stress alert per animal and per barn

Mitigation — cheapest first



Shade: 3.5–4.5 m² per adult, roof height $\geq$ 4 m, open sides



Water: unlimited, cool, $\geq$ 10 cm trough per animal; buffalo wallow or sprinkler



Fans + soaking cycles at feed line and holding area when THI > 72



Feed 60–70% of the ration in the evening/night; fresh feed, more forage quality, less heat of fermentation



Breed in the cooler months where possible; use timed AI; expect lower conception in peak heat

Biosecurity — keep it out, keep it in, keep records



Keep it out

- Single controlled entry; visitors and vehicles disinfected
- Quarantine new animals 30 days, test before mixing
- Buy from known herds; check vaccination and test papers
- Feed and bedding from clean sources; rodent and bird control



Keep it in (contain)

- Isolate sick animals; treat and milk them last
- Footbaths and separate tools for the sick pen
- Safe carcass disposal; never sell sick animals into the market
- Report notifiable diseases (FMD, HS, anthrax, brucellosis, TB) immediately



Programme

- Vaccination per DA-BAI / PCC schedule: FMD, haemorrhagic septicaemia, others as advised
- Strategic deworming and tick/fly control; monitor with faecal counts
- Brucellosis and TB testing where the programme applies
- Every vaccination, test and treatment entered in iHealth

Records and responsible drug use make data useful

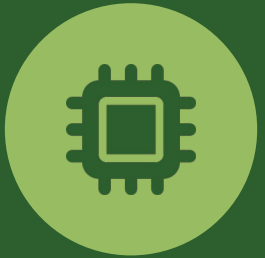
Minimum record set per animal (NDHIS / iHealth)

Event	Fields
Identification	ID / ear tag, breed, birth date, dam, sire
Reproduction	Heat date, AI date + sire + technician, PD result + date, calving date + ease + calf ID
Health	Diagnosis, treatment, product, dose, route, dates, withdrawal end date, outcome
Production	Milk yield per test day, SCC/CMT, body condition, weight
Movements	Purchase, sale, death with reason

A record you cannot find in one minute does not exist. Enter on the day, on the phone.

Golden rules for veterinary drugs

- ✓ Diagnosis first — a thermometer and CMT before a syringe
- ✓ Only products registered for the species; prescription products only under the veterinarian
- ✓ Dose by body weight (weigh tape), correct route (IM / SC / IV / IMM), full course
- ✓ One sterile needle per animal; clean injection sites; separate IMM tubes
- ✓ Withdrawal period: mark the animal, record start and end dates, keep milk out of the tank
- ✓ Store cool, dark, dry; check expiry; never mix products in one syringe
- ✓ Record every treatment the same day — this is also your legal protection



MODULE 3

Digital herd monitoring

Rumen bolus sensors: what they measure, the alerts they generate, real cases, integration with NDHIS / iHealth / iFeed, and how to design the Philippine pilot.

From eyes to data — closing the observation gap

What a technician cannot see

- Heat at 2 a.m. in a buffalo that shows almost nothing at 8 a.m.
- A fever that starts 2–4 days before the cow looks sick
- Rumination that stopped this morning — the cow still stands at the feed bunk
- A calving that begins tonight in a pen nobody is watching
- Which of 50 animals is actually suffering from the heat
- A herd-wide temperature rise = infection or vaccine reaction

What the animal tells us, continuously



Core body temperature

fever, calving, heat stress, drinking events



Rumination (reticular contractions)

health, feed intake, acidosis, transition



Activity

estrus, lameness, illness



Water intake events

first sign of many diseases



Rumen pH (optional sensor)

acidosis, ration changes

Sensor technologies — what each can and cannot measure

	Neck collar	Leg pedometer	Ear tag	Rumen bolus
Measures	Activity, rumination (by neck movement)	Activity, lying time	Activity, ear-surface temp	Core temperature, rumination (direct), activity, water intake, optional pH
Temperature	No	No	Indirect (skin, weather-affected)	Direct, ± 0.05 °C
Rumination	Indirect	No	Indirect	Direct — counts reticular contractions
Loss / damage	Strap loss, needs re-fitting as animal grows	Loss in mud, wallow	Tears out, loss	Inside the animal; tamper-proof; no maintenance
Battery / life	2–5 yrs, recharge or replace	2–5 yrs	2–3 yrs	≥ 4 years, no maintenance
Buffalo suitability	Wallowing damages straps	Mud and water	Ear damage	Unaffected by wallowing — needs local validation of thresholds

No single device is perfect: collars and tags are cheaper to buy; the bolus measures more, directly, for longer — and is the only option that survives a buffalo wallow.

How a rumen bolus works



1 · Bolus

Given orally with an applicator; settles in the reticulum for life



2 · Base station

Receives radio data in the barn; optional climate (THI) sensor



3 · Cloud & AI

Self-learning models per animal; 7-day onboard buffer if offline



4 · App / dashboard

Web, iOS, Android — alerts to farmer, technician, veterinarian

How it reads rumination

The bolus sits in the reticulum and counts its contractions. During rumination the reticulum contracts three times per cycle; while eating or resting, twice. So the sensor knows — directly, not by inference — how many minutes a day the animal ruminates, and alerts when it falls.

Device facts (smaXtec bolus used in Korea)

- Basic: 105 × 35 mm; temperature 0-50 °C, ± 0.05 °C; battery ≥ 4 years; bio-polymer housing
- Premium: 135 × 35 mm, adds rumen pH (3-9) for ~150 days
- No maintenance, no re-fitting, cannot be lost or swapped between animals
- Requires adult animal (≥ ~300 kg) and a trained person with the applicator

The alerts a technician receives



Reproduction

- Heat detected + optimal insemination window
- Calving imminent (typically 10–20 h ahead)
- Post-calving recovery watch (fever, rumination)



Health

- Temperature rise (fever) — up to ~4 days before clinical signs
- Rumination drop · activity drop
- Suspected ketosis · milk fever · mastitis
- Reduced water intake · reduced feed intake



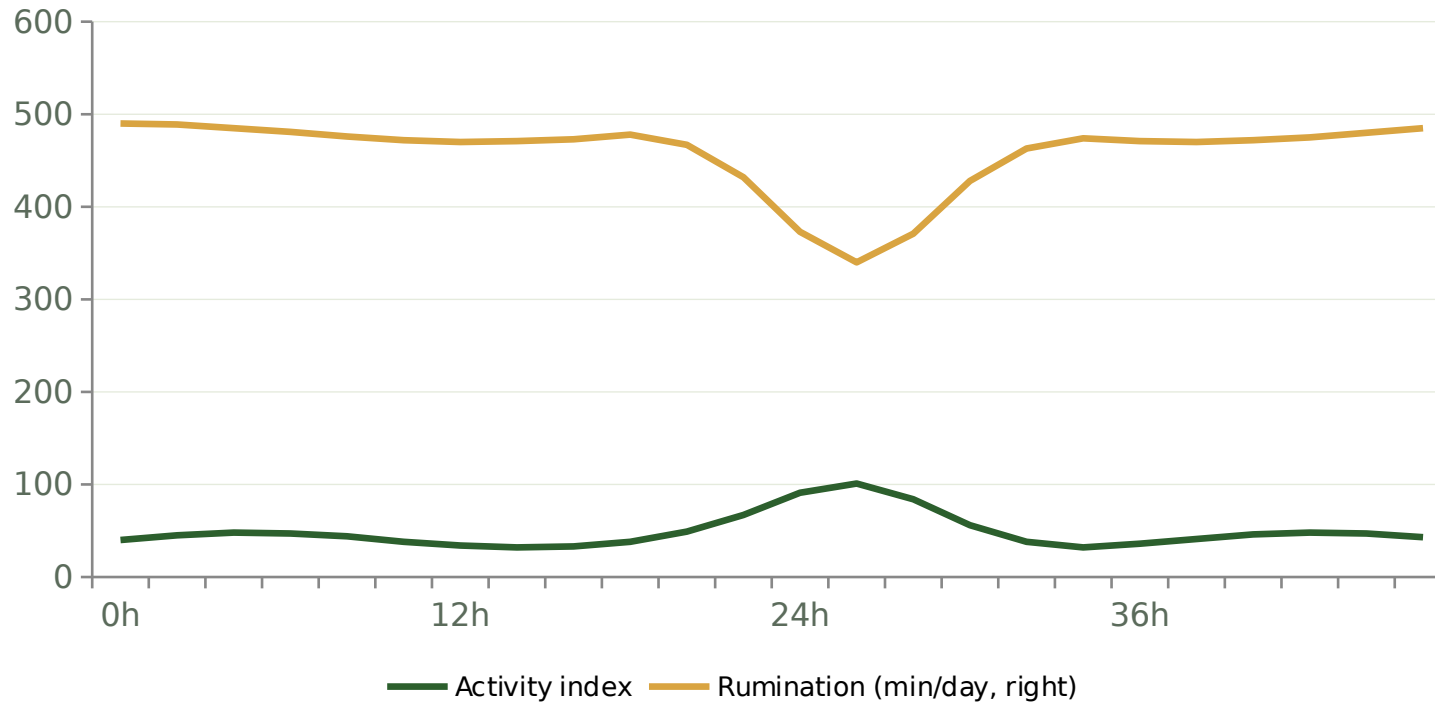
Feeding & environment

- Heat-stress alert per animal and per barn (THI)
- Ration change effects on rumination (and pH with premium bolus)
- Group-level trends for feeding management

Performance figures (heat detection > 90%, calving detection > 90%, illness alerts days before clinical signs, antibiotic reduction up to 70%) are manufacturer and user reports from cattle herds. They are targets to verify in the Philippine pilot, not guarantees.

Case 1 · Heat alert — the activity curve tells you when

Illustrative 48-hour trace of one cow: activity up, rumination down = heat



Illustrative curve, not measured data. Real alerts also use temperature and the animal's own baseline.

What the technician does

- Alert at 20:00 with onset time; app shows the insemination window (next morning)
- Confirm on farm: mucus, tail-paint, standing — or trust the sensor for silent heats
- Inseminate inside the window; record AI + sire in NDHIS
- Next expected heat and pregnancy-check date are generated automatically
- Reported outcomes in cattle herds: fewer missed heats, days open reduced by up to ~25% (user reports)

Case 2 · Calving alert and the fresh-cow watch

Illustrative: temperature drop before calving (0 h), then a fever rising from +48 h (metritis)



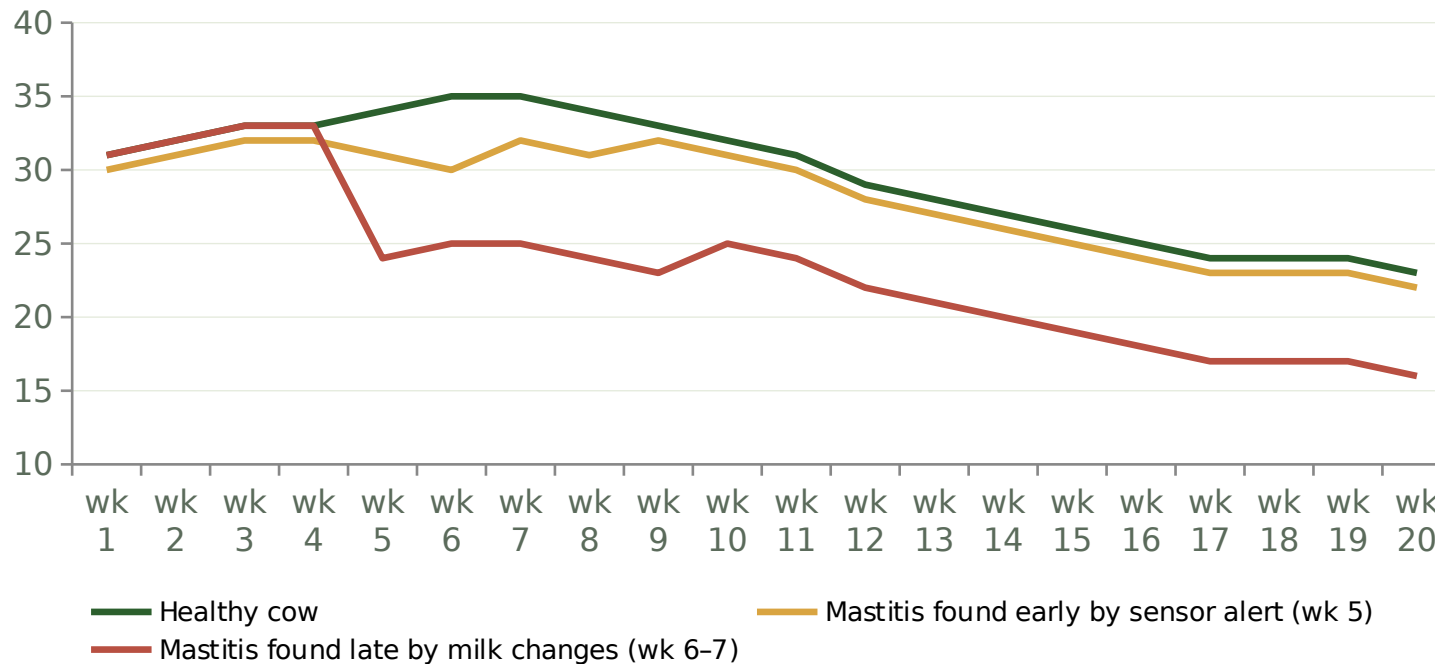
Illustrative curve. A 2017 performance study at Georg-August-University Göttingen reported most calving alerts 10–20 h before calving (manufacturer citation).

Action list

- Calving alert → move to clean pen, prepare colostrum, plan the night check
- No progress alerts? Still apply the 30 min / 2 h rules from Module 1
- Days 1–14: temperature and rumination watch — metritis, ketosis and mastitis show here first
- Fever alert on day 3 → examine, CMT, discharge check → treat per protocol
- Enter calving ease, calf ID and every treatment in NDHIS / iHealth

Case 3 • Early mastitis detection saves the lactation

Illustrative milk yield (kg/day) after calving — the earlier the treatment, the smaller the loss



Illustrative curves based on the pattern reported by smaXtec users; actual losses vary with pathogen and treatment.

Why early matters

- Temperature and rumination change 1–4 days before milk looks abnormal (reported)
- Early case: often mild → strip, NSAID, monitor; fewer antibiotics, shorter withdrawal
- Late case: acute quarter, systemic signs, permanent tissue damage, lower peak
- Each early case saves discarded milk, drugs and — most of all — the rest of the lactation
- Combine: sensor alert → CMT → decision → record

Case 4 · Herd-level signals across the whole farm



Vaccination reaction

Many animals show a temperature rise 1–2 days after vaccination. Normal — but it tells you the vaccine was given and how the herd responded.



Infectious outbreak

A temperature rise across many animals with no vaccination = investigate today. Early warning for FMD, HS and respiratory disease.



Heat stress day

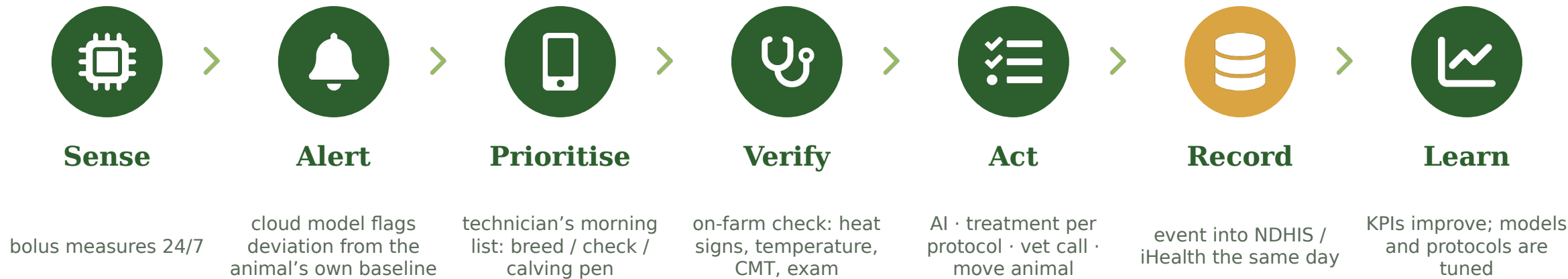
Body temperature and water-intake alerts cluster in the afternoon; rumination falls at night. Measure whether fans and sprinklers actually work.



Feed change

A ration change shows in rumination within 24 h — and in rumen pH with the premium bolus. Catch acidosis before hooves and milk fat show it.

From alert to action to record — the daily loop



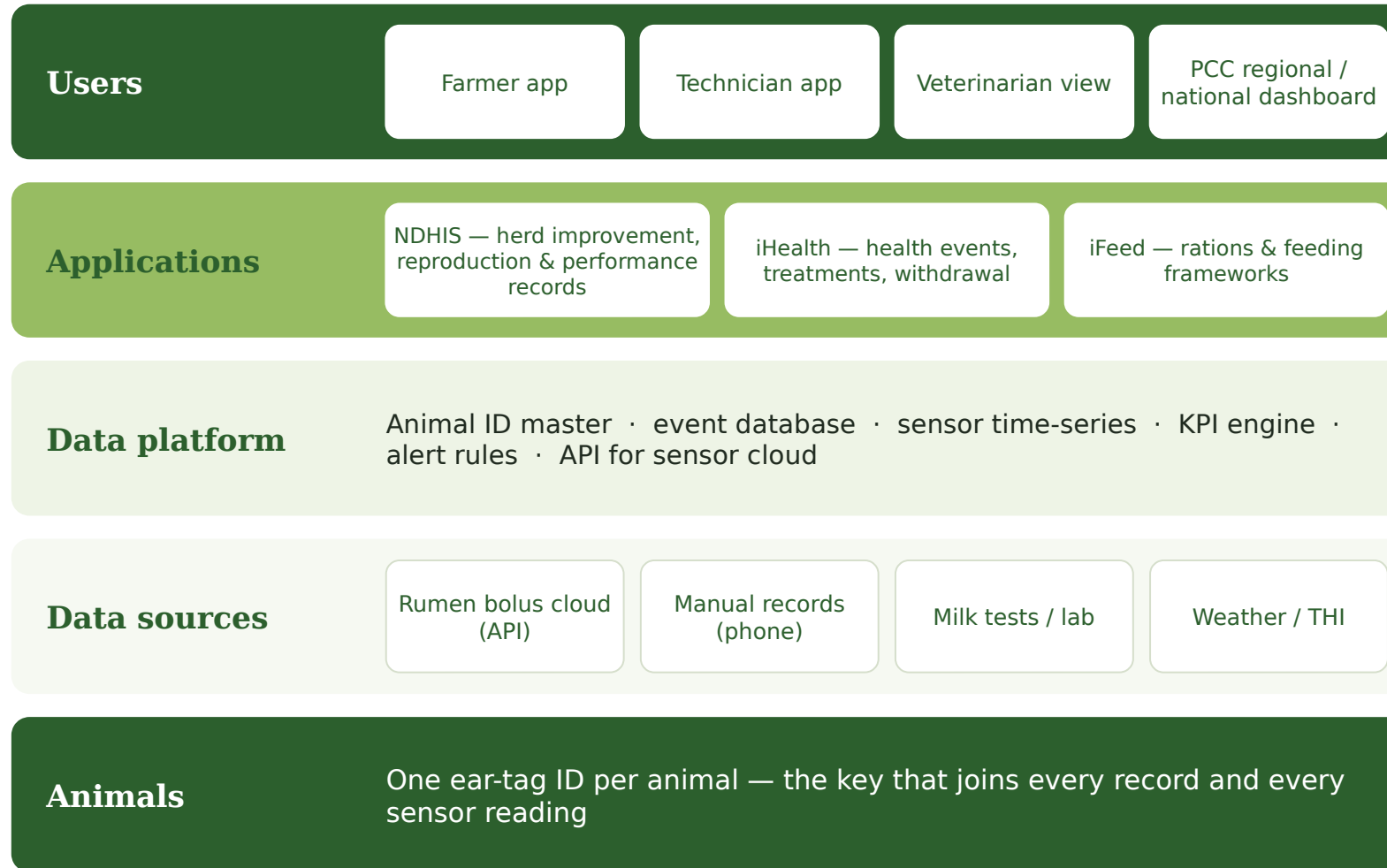
Who does what

Role	Daily responsibility
Farmer	Sees own animals' alerts; feeds, moves, calls technician
Livestock technician	Morning list, AI, PD, CMT, first-line treatment, data entry
Veterinarian	Protocols, severe cases, herd health review of alert trends
PCC regional / national	Dashboards, KPIs, outbreak signals, programme decisions

Rules that keep the loop alive

- Every alert gets a response in the app — even “checked, nothing found”
- Record on the day; unrecorded action = lost learning
- Weekly 30-minute review: alerts vs outcomes, per farm
- Adjust thresholds per animal group (buffalo vs cattle, heifers vs cows) with the veterinarian

Fitting sensors into NDHIS, iHealth and iFeed



Korea's approach: "CowTalk AI"

- Sensor data + public livestock data + AI models in one platform
- Same data serves farmer, vet, inseminator and government — different views
- Designed as a provincial livestock operating system: farm → province → national
- Sensor cloud connected by API, so NDHIS stays the system of record

Designing the Philippine pilot with 100 sensors

Buffalo-specific points

- Reticulum anatomy and bolus retention are comparable to cattle — confirm with the first 10 animals (ultrasound / X-ray optional)
- Baseline temperature, rumination minutes and activity differ from Holsteins → collect 4–6 weeks of baseline before trusting alerts
- Wallowing lowers body temperature transiently — models must learn this pattern
- Heat signs weak → the heat alert is the primary target outcome
- Applicator technique and restraint: trained staff only

Pilot design (proposal for discussion)

Element	Proposal
Animals	100 sensors: e.g. 60 dairy buffalo + 40 dairy cattle across 4–6 cooperative or institutional farms; adult animals in production
Comparison	Paired control animals on the same farms without sensors (or before/after per animal) — the Korean Gyeonggi pilot uses this RCT-style design
Baseline	Months 1–2: fit sensors, collect baseline, train technicians; no alert-based actions yet
Intervention	Months 3–12: alert-driven heat detection, fresh-animal monitoring, disease checks; all events in NDHIS / iHealth
Primary KPIs	Heat detection rate · conception rate · days open · calving alerts confirmed · disease detection lead time · antibiotic treatments per 100 animal-months
Secondary	Milk yield, calf mortality, technician time per animal, farmer satisfaction, data completeness in NDHIS
Deliverable	Month 12 report: measured effects, buffalo thresholds, cost per animal-month, recommendation for scale-up

Where the money comes from — value drivers for the pilot

Value driver	Mechanism	How to measure it	Direction reported in cattle herds
Fewer days open	Heats caught, AI timed correctly	Days open before vs after; per-day cost of open animal (feed + lost milk)	Down (up to ~25% reported)
Fewer clinical mastitis cases, milder cases	Early temperature / rumination alert → early CMT	Cases per 100 animal-months; discarded milk; drug cost	Down; antibiotic use down (up to ~70% reported)
Fewer calving losses	Calving alert → supervised calving	Stillbirths, dystocia, retained placenta per 100 calvings	Down
Earlier treatment of metabolic disease	Fresh-animal watch	Ketosis / milk fever cases, culls in first 60 days	Down
Labour efficiency	Morning list replaces heat watching and thermometer rounds	Technician hours per animal per week	Down
Milk yield	Sum of the above	kg per animal per lactation, adjusted for season	Up (varies)

Simple pilot economics (fill with local numbers): value per animal-month = Δ days open $\times$ cost/day + Δ mastitis cases $\times$ cost/case + Δ calf losses $\times$ calf value + labour saved $\times$ wage. Compare with sensor cost per animal-month (device amortised over 4 years + platform fee). Korean dairies typically recover the cost from reproduction alone; buffalo economics must be measured, not assumed.

Reference cases: farm → province → international



Song Young Shin Farm, Korea

120 head · 90 bolus sensors · robots · compost-bedded barn. Night heat watching and fresh-cow thermometer rounds replaced by phone alerts; first animal-welfare and low-carbon certified dairy in Korea.



Gyeonggi Province AI livestock platform (planned)

Five-year provincial programme; dairy ICT pilot designed for 31 municipalities × 5 farms × 50 milking cows (7,750 cows) with randomised comparison; rumen sensors specified as the data source; expansion to beef and pig farms.



Uzbekistan pilot (≈ 50 cows)

First international validation: sensor-based monitoring on a dairy herd with government stakeholders; paired with Korean Holstein sexed-embryo programme run with the Rural Development Administration (RDA). Figures being finalised.

The lesson: the same sensor data serves the farmer, the veterinarian and the government — design the pilot so it can scale to the region and the nation.



MODULE ★

Wrap-up and practicum

Ten things to take back to your farms, and this afternoon's hands-on stations.

Ten things to take back to your farms

- 1 Identify every animal and record every event — nothing else works without this
- 2 Heat detection rate is the first KPI to fix; silent and night heats are the main loss
- 3 AI 8–16 h after heat onset; treat semen like a patient
- 4 Ultrasound pregnancy check at day 28–30; re-enrol open animals the same day
- 5 Calving: 30 min / 2 h rules; colostrum 4 Qs within 2 hours
- 6 Transition cows: keep them eating — rumination is the early warning
- 7 Mastitis: CMT, treat by grade and protocol, respect withdrawal, post-dip every teat
- 8 Heat stress: shade, water, air, night feeding — measure it per animal
- 9 Sensors close the observation gap: alert → verify → act → record, every day
- 10 Pilot with a baseline and a comparison group so the results can drive national scale-up

Practicum stations — this afternoon



Station 1 • Heat detection & AI timing

Tail paint, mucus and standing-heat scoring on live animals; semen thawing and gun loading; AM/PM decision cards

Kit: Heat detection aids, semen tank, thaw bath, AI kit



Station 2 • Pregnancy diagnosis

Rectal ultrasound at day 28–60: finding the uterus, embryo, heartbeat; palpation basics on buffalo

Kit: Portable ultrasound, gloves, lubricant, restraint chute



Station 3 • Mastitis & milking routine

CMT on quarters, grading, treatment-decision cards; pre-dip / dry / post-dip demonstration

Kit: CMT paddle & reagent, dips, towels, sample tubes



Station 4 • Rumen bolus & app

Bolus and applicator handling, safe administration on an adult animal, base-station check, reading alerts and entering the response

Kit: Bolus, applicator, base station, phone/tablet with app



Station 5 • NDHIS / iHealth data entry

Enter the day's heat, AI, PD, CMT and treatment events; view the KPI screen

Kit: Tablets, NDHIS training accounts



Station 6 • Fresh-cow & calf check

Temperature, rumen fill, BCS, navel and scour assessment; colostrum Brix reading

Kit: Thermometer, Brix refractometer, BCS chart, colostrum

Maraming salamat po.

Healthy soil → healthy grass → healthy cow → healthy milk → healthy people.

The data just helps us see it sooner.

Dr. Hyunje Ha, DVM

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Questions, farm data and pilot design: through the PCC-KOICA NDHIP project office

